

A Complete and Natural Rule Set for Multi-Qudit Clifford Circuits in All Odd Prime Dimensions

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This talk

- A **presentation by generators and relations** of the n -qudit Clifford group, for every odd prime dimension p : 16 relations, none involving more than 3 qudits.
- A **unique normal form** for symplectic (= Clifford modulo Pauli) circuits.
- All, except for two math facts, are **formalised in Agda**.

About this slides

- Every Agda code block is typechecked by Agda.
- Written by *Claude* and Xiaoning Bian (~200-line outline prompt and slides checking & improvement).

Introduction

From qubit to qudit Clifford gates

Fix an odd prime p and let $\omega = e^{2\pi i/p}$. The Clifford group $\mathcal{C}_n$ is generated by:

	qubit ($p = 2$)	qudit (odd prime p)
H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$ j\rangle \mapsto \frac{1}{\sqrt{p}} \sum_k \omega^{jk} k\rangle$
S	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$ j\rangle \mapsto \omega^{j(j-1)/2} j\rangle$
CZ	$\text{diag}(1, 1, 1, -1)$	$ j\rangle l\rangle \mapsto \omega^{jl} j\rangle l\rangle$

Note the $/2$ in the qudit S is the inverse of 2 in $\mathbb{Z}_p$. A global phase can be added to the definition of H to facilitate computation.

The setting: one Agda module

These slides are a single Agda module, parameterised by the prime p and by a primitive root g of $\mathbb{Z}_p^*$:

```
module qpl26slides.talk
  (p-3 : ℕ)
  (let p-2 = 1 + p-3)
  (p-prime : Prime (3 + p-3))
  (let open PrimeModulus' p-2 p-prime)
  (g*@ (g , g≠0) : ℤ* p)
  (g-gen : ∀ ((x , _) : ℤ* p) → ∃ \ (k : ℤ p-1) → x ≡ g ^ toℕ k)
  where
```

- $p = p-3 + 3$: every odd prime, uniformly.
- Everything downstream — rule sets, normal forms, theorems — is parameterised by the choice of g .

Circuits are words

A gate set is a family of gate types indexed by arity; a circuit on n wires is a **word** over wire-indexed generators.

```
data SympGate :  $\mathbb{N} \rightarrow$  Set where
```

```
  H-gate : SympGate 1
```

```
  S-gate : SympGate 1
```

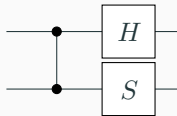
```
  CZ-gate : SympGate 2
```

```
example : Circuit 2
```

```
example = CZ • H ↑ • S
```

- `Circuit  $n$  = Word (Gen  $n$ )`: free monoid; `↑` shifts a circuit up one wire.

- diagrams read in matrix-multiplication order: `example =`



Derived gates: the rest of the usual gate set

X , Z , the swap Ex , and the two controlled- X gates are *words*, not extra generators:

$$ZX : \forall \{n\} \rightarrow \text{Word} (\text{Gen} (1 + n))$$

$$Z = H \bullet H \bullet S \bullet H \bullet H \bullet S^{-1}$$

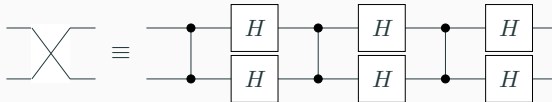
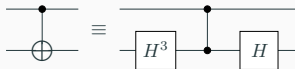
$$X = H \bullet S \bullet H \bullet H \bullet S^{-1} \bullet H$$

$$Ex \text{ CX } XC : \forall \{n\} \rightarrow \text{Word} (\text{Gen} (2 + n))$$

$$Ex = CZ \bullet H \downarrow \bullet H \uparrow \bullet CZ \bullet H \downarrow \bullet H \uparrow \bullet CZ \bullet H \downarrow \bullet H \uparrow$$

$$CX = H \downarrow \wedge^3 \bullet CZ \bullet H \downarrow$$

$$XC = H \uparrow \wedge^3 \bullet CZ \bullet H \uparrow$$



Main results

Main result 1: a presentation of the qudit Clifford group

The paper's Figure 1 rule set, mod scalars, **presents** the projective Clifford group

$$\mathcal{C}_n / \langle \omega \rangle \cong \text{Pauli}_n \rtimes \text{Sp}(2n, \mathbb{Z}_p):$$

`Theorem-1 : ∀ (n : ℕ) → (n CRel, _==_) IsPresentationOf (Pauli×Sp n)`

`Theorem-1 = PapPres.presentation`

- `_CRel, _==_` is the 16-relation rule set, shown on the next slides.
- The theorem holds *at every width n*, for *every* odd prime p and every primitive root g — the whole deck is parameterised by them.

The 16 relations, in Agda (one qudit)

```
data _CRel, _===_ : (n : ℕ) → WRel (Gen n) where

order-S      :      (1+ n) CRel, S ^ p === ε
order-H      :      (1+ n) CRel, H ^ 2 === M-1
M-power      : ∀ k → (1+ n) CRel, Mg ^ k === M (gk)
semi-MR      :      (1+ n) CRel, R • Mg === Mg • Rg (g * g)
order-SH     :      (1+ n) CRel, (S • H) ^ 3 === ε
comm-HHSHHS  :      (1+ n) CRel, H • H • S • H • H • S === S • H • H • S • H • H
```

- M_x is the multiplier $|j\rangle \mapsto |xj\rangle$, spelled as the word $R^{x^{-1}} H R^x H R^{x^{-1}} H$ with $R = S \cdot Z^{1/2}$.
- One-qudit relations talk about the powers of a **fixed primitive root** g — this is where odd primes differ from the qubit case.

The 16 relations, in Agda (two qudits)

`order-CZ` : $(z + n) \text{ CRel}, CZ^p \equiv \epsilon$

`order-Ex` : $(z + n) \text{ CRel}, Ex^2 \equiv \epsilon$

`comm-CZ-S†` : $(z + n) \text{ CRel}, CZ \cdot S^\dagger \equiv S^\dagger \cdot CZ$

`semi-M†CZ` : $(z + n) \text{ CRel}, CZ \cdot Mg^\dagger \equiv Mg^\dagger \cdot CZ^g$

`semi-Ex-S†` : $(z + n) \text{ CRel}, Ex \cdot S^\dagger \equiv S^\dagger \cdot Ex$

`semi-Ex-H†` : $(z + n) \text{ CRel}, Ex \cdot H^\dagger \equiv H^\dagger \cdot Ex$

`blake-c12` : $(z + n) \text{ CRel}, (S^{p-1})^\dagger \cdot (S^{p-1})^\downarrow \cdot CX^{p-1} \cdot S^\downarrow \cdot CX \equiv CZ$

- `blake-c12` converts between the two entangling gates CZ and CX .

The 16 relations, in Agda (three qudits)

`yang-baxter` : $(\exists + n) \text{ CRel}, \text{Ex } \uparrow \bullet \text{Ex } \downarrow \bullet \text{Ex } \uparrow === \text{Ex } \downarrow \bullet \text{Ex } \uparrow \bullet \text{Ex } \downarrow$

`cz-slide` : $(\exists + n) \text{ CRel}, \text{Ex } \downarrow \bullet \text{Ex } \uparrow \bullet \text{CZ} === \text{CZ } \uparrow \bullet \text{Ex } \downarrow \bullet \text{Ex } \uparrow$

`semi-CX $\uparrow$ -CZ $\downarrow$` : $(\exists + n) \text{ CRel}, \text{CZ } \downarrow \bullet \text{CX } \uparrow === \text{CZ02} \bullet \text{CX } \uparrow \bullet \text{CZ } \downarrow$

- Only **these 16** are axioms. $XZ = ZX \pmod{\omega}$, *Ex*-vs-*CZ*, and both Pauli-vs-*CZ* rules are *derived*.
- Plus, for every gate set, the same three monoidal structural rules: congruence under $\uparrow$, gates on disjoint wires commute, and scalars commutes with every gate.

The 16 relations, as circuits (1-2 qudits)

order-S $\begin{array}{c} \text{---} \boxed{S^p} \text{---} \end{array} \equiv \text{---}$

order-H $\begin{array}{c} \text{---} \boxed{H} \boxed{H} \text{---} \end{array} \equiv \begin{array}{c} \text{---} \boxed{M_{-1}} \text{---} \end{array}$

M-power $\begin{array}{c} \text{---} \boxed{(M_g)^k} \text{---} \end{array} \equiv \begin{array}{c} \text{---} \boxed{M_{g^k}} \text{---} \end{array}$

semi-MR $\begin{array}{c} \text{---} \boxed{M_g} \boxed{R^{g^2}} \text{---} \end{array} \equiv \begin{array}{c} \text{---} \boxed{R} \boxed{M_g} \text{---} \end{array}$

order-CZ $\begin{array}{c} \text{---} \bullet \\ | \\ \text{---} \bullet \end{array} \equiv \text{---}$

order-Ex $\begin{array}{c} \text{---} \diagup \diagdown \diagup \diagdown \text{---} \end{array} \equiv \text{---}$

comm-CZ-S $\begin{array}{c} \bullet \text{---} \boxed{S} \\ | \\ \bullet \text{---} \end{array} \equiv \begin{array}{c} \boxed{S} \text{---} \bullet \\ | \\ \bullet \text{---} \end{array}$

semi-M $\uparrow$ CZ $\begin{array}{c} \boxed{M_g} \text{---} \bullet \\ | \\ \bullet \text{---} \end{array} \equiv \begin{array}{c} \bullet \text{---} \boxed{M_g} \\ | \\ \bullet \text{---} \end{array}$

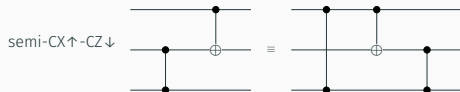
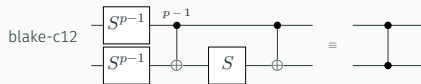
semi-Ex-S $\uparrow$ $\begin{array}{c} \diagup \diagdown \diagup \diagdown \text{---} \boxed{S} \end{array} \equiv \begin{array}{c} \boxed{S} \text{---} \diagup \diagdown \diagup \diagdown \end{array}$

semi-Ex-H $\uparrow$ $\begin{array}{c} \diagup \diagdown \diagup \diagdown \text{---} \boxed{H} \end{array} \equiv \begin{array}{c} \boxed{H} \text{---} \diagup \diagdown \diagup \diagdown \end{array}$

order-SH $\begin{array}{c} \text{---} \boxed{S} \boxed{H} \boxed{S} \boxed{H} \boxed{S} \boxed{H} \text{---} \end{array} \equiv \text{---}$

comm-HHSHHS $\begin{array}{c} \text{---} \boxed{H} \boxed{H} \boxed{S} \boxed{H} \boxed{H} \boxed{S} \text{---} \end{array} \equiv \begin{array}{c} \text{---} \boxed{S} \boxed{H} \boxed{H} \boxed{S} \boxed{H} \boxed{H} \text{---} \end{array}$

The 16 relations, as circuits (2–3 qudits)



What “presents” means

```
record _Presents_ ( _===_ : WRel X) (G : Group a ℓ) : Set (a ⊔ ℓ) where
  field gl : Grouplike _===_
  module GL = Group-Lemmas _===_ gl
  open GroupMorphisms (Group.rawGroup GL.⋅-ε-group) (Group.rawGroup G)
  field
    [ ] : Word X → Group.Carrier G
    iso : IsGroupIsomorphism [ ]
```

$$\text{Cir } n / \approx_{16 \text{ rules}} \xrightarrow[\cong]{[\]} \text{Pauli}_n \rtimes \text{Sp}(2n, \mathbb{Z}_p) \cong \mathcal{C}_n / \langle \omega \rangle$$

- *Soundness*: $[\]$ is well defined — each rule holds semantically.
- *Completeness*: $[\]$ is injective — semantically equal circuits are related by the rules.

Main result 2: a unique normal form for symplectic operators

Symplectic operators are Clifford mod Pauli: the “proper” part. Normal forms are **data**, built from four atomic boxes:

A B D E : Set

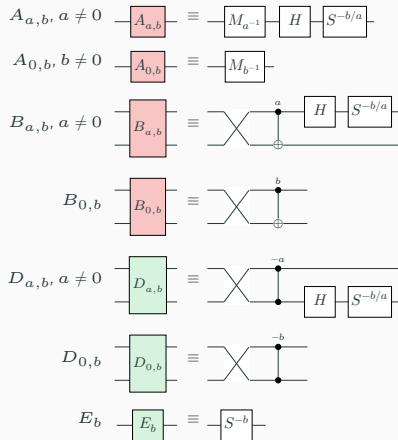
$A = \Sigma[ab \in (\mathbb{Z}_p \times \mathbb{Z}_p)]$

$(ab \neq (0, 0))$

$B = \mathbb{Z}_p \times \mathbb{Z}_p$

$D = \mathbb{Z}_p \times \mathbb{Z}_p$

$E = \mathbb{Z}_p$



Rows, and the staircase

$$M \ L \ ML : \mathbb{N} \rightarrow \text{Set}$$

$$M \ 0 = T$$

$$M \ (1 + n) = \text{Vec } D \ n \times E$$

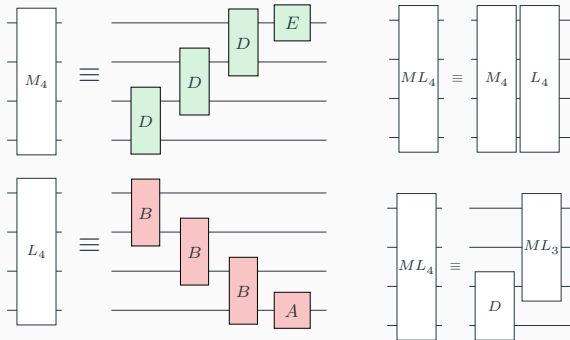
$$L \ 0 = T$$

$$L \ (1 + n) = \text{Vec } B \ n \times A$$

$$ML \ 0 = T$$

$$ML \ 1 = M \ 1 \times L \ 1$$

$$ML \ (2 + n) = M \ n \times L \ n \cup D \times ML \ (1 + n)$$



A double induction: the rows M , L and the boxes ML all recurse on the width.

The normal form, as a circuit

Reading normal forms back as circuits is structural recursion:

$NF : \mathbb{N} \rightarrow \text{Set}$

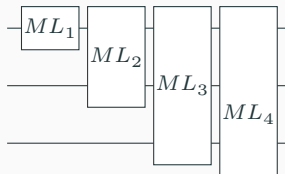
$NF\ 0 = \top$

$NF\ (1 + n) = NF\ n \times ML\ (1 + n)$

$[]^{nf} : \forall \{n\} \rightarrow NF\ n \rightarrow \text{Word}\ (\text{Gen}\ n)$

$[]^{nf}\ \{0\}\ \text{tt} = \varepsilon$

$[]^{nf}\ \{1 + n\}\ (nf, ml) = [nf]^{nf}\ \uparrow \bullet [ml]^{ml}$



Everything on these slides is Agda-checked

- ≈ 320 files, $\approx 100\,000$ lines of Agda, all `--safe`, *no postulates*.
- Agda 2.8, standard library 2.3.
- One root (`MainTheorems.agda`) reaches the whole library; these slides import the library and *re-state* the main theorems — if a statement on a slide did not match the formalisation, the slides would not build.

What the formalisation covers

All group-theoretic computation: presentations, normal forms, completeness. The two glue facts — that the projective Clifford group is $\text{Pauli} \rtimes \text{Sp}$, and that the Clifford group is a central extension of it by $\langle \omega \rangle$ — are classical and are *not* formalised.

Method

Divide and conquer

Factor the group; present the factors; compose the presentations.

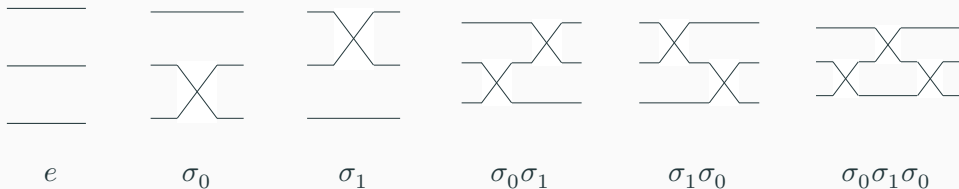
$$1 \rightarrow \text{Pauli}_n \rightarrow \mathcal{C}_n / \langle \omega \rangle \rightarrow \text{Sp}(2n, \mathbb{Z}_p) \rightarrow 1 \quad \text{semidirect: it splits}$$

$$1 \rightarrow \langle \omega \rangle \rightarrow \mathcal{C}_n \rightarrow \mathcal{C}_n / \langle \omega \rangle \rightarrow 1 \quad \text{central extension}$$

- Doesn't this just move the difficulty around? **No**: the symplectic factor is where the real rewriting happens, and keeping the Pauli part out of it pays off everywhere.
- Prior work (qubit: Selinger; qutrit) worked with the whole Clifford group at once — the derivations drag a “messy” Pauli correction through every step.

Warm-up: the symmetric group S_n

Swap circuits on n wires present S_n . All of S_3 :



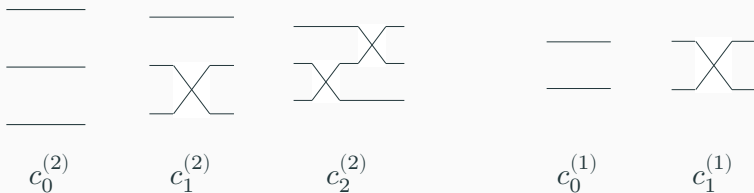
Circuits have a built-in tower of subgroups

$$\text{Cir } 0 < \text{Cir } 1 < \dots < \text{Cir } n$$

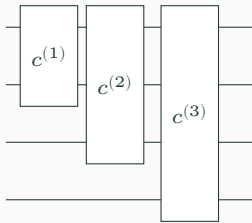
($\text{Cir } k$ = circuits that only touch the top k wires). For S_n : $S_0 < S_1 < \dots < S_n$.

Coset representatives, and the normal form

Representatives of S_3/S_2 (and of S_2/S_1): the staircases



- For $f \in S_3$ there is a unique $c \in C_2$ with $f^{-1}c$ fixing the bottom wire: c must send $f^{-1}(0)$ to 0.
- Recurse on $f^{-1}c \in S_2$: every f is uniquely $c^{(2)} \cdot (c^{(1)} \uparrow)$.
- Group theory: given $G_0 < \dots < G_n$ and coset reps C_k of G_k/G_{k-1} , elements of G_n factor uniquely as $\prod_k c_k$ — a **mixed-radix encoding**: $|S_3| = 3 \cdot 2 \cdot 1$.



Rewriting to the normal form: push a generator through

Multiply a normal form by a generator; push it through the staircase:



disjoint wires: commute past



Yang-Baxter: pass through, shift up



$\sigma^2 = e$: staircase shrinks

From S_n to $\text{Sp}(2n, \mathbb{Z}_p)$: change the action

The same framework, three ways of looking at the wires:

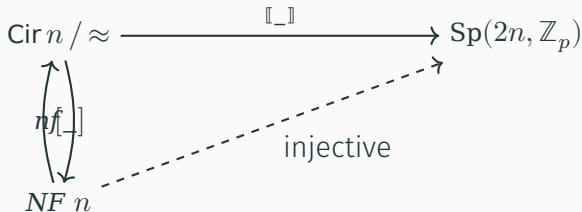
1. S_n acts on $\{0, \dots, n-1\}$: a wire is a *position*; c sends $f^{-1}(0)$ to 0.
2. S_n acts on **Vec Bit** n : a wire is a *bit*; $f^{-1}c$ fixes the linear subspace of the bottom wire.
3. Adding H, S, CZ : the group acts on **Vec Pauli₁** n — a $2n$ -dimensional **symplectic** $\mathbb{Z}_p$ -vector space. A wire is a **Pauli₁**; enlarging the coset layer from staircases to **ML** boxes, $f^{-1}c$ fixes the symplectic subspace of the bottom wire.

The framework applies to *any* $\mathbb{N}$ -indexed family of groups with such a layered structure (**Circuit.CosetNF**, **Circuit.Uniqueness**).

Uniqueness, in Agda

Theorem-2 : $\forall n \{u \ v : \text{NF } n\} \rightarrow \llbracket [u] \rrbracket \approx^s \llbracket [v] \rrbracket \rightarrow u \equiv v$

Theorem-2 = U. $\llbracket [] \rrbracket$ -injective



Soundness + unique normal forms $\Rightarrow$ completeness (**by-normalization**); the interesting content is that the *coset factor is determined by the action* on Pauli states.

From 66 relations to 18

1. Rewrite *every* circuit to normal form; collect every relation the rewrite ever uses: **66 relations** (box-vs-gate commutations, case by case).
2. By inspection, find a small set of **primitive, bidirectional** relations that derive all 66.
3. The choice is not canonical — pick the set you find natural.

An example of a derived rule, $M_x \cdot M_y = M_{xy}$:

$$\boxed{M_x} \boxed{M_y} \equiv \boxed{M_{xy}}$$

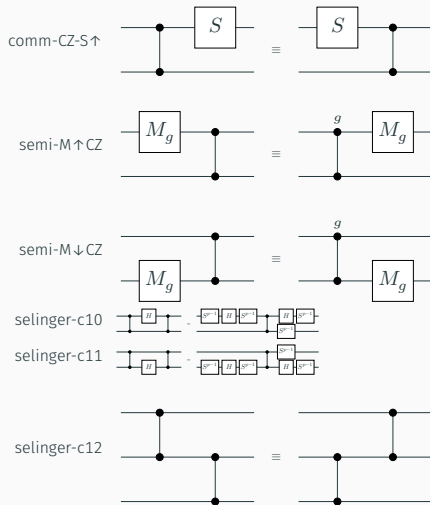
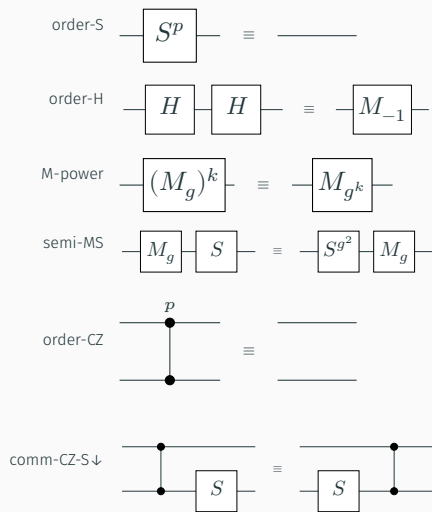
```
M-mul : ∀ n (x y : ℤ* p) → let open PB (Sim._QRel, _===_ (1 + n)) in
```

```
  M x • M y ≈ M (x *' y)
```

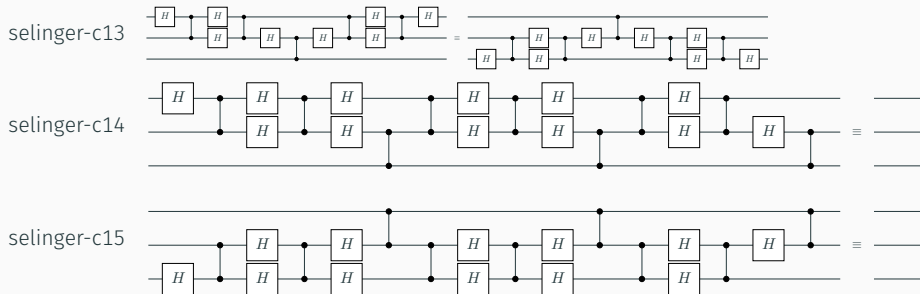
```
M-mul n = SimL.Lemmas1.lemma-M-mul n
```

$(x = g^k, y = g^l; \mathbf{M}\text{-power}$ twice; add exponents mod $p - 1$.)

The simplified symplectic rule set (1-2 qudits)



The simplified symplectic rule set (3 qudits), and the theorem



Theorem-Sp : $\forall \{n\} \rightarrow (n \text{ SRel}, _ == _) \text{ IsPresentationOf } (\text{Sp-group } n)$

Theorem-Sp = SimPres.presentation

Comparison with the qubit rule set (Selinger)

- **identical** 3-qudit relations (C12–C15);
- **almost identical** 2-qudit relations;
- **more complicated** 1-qudit relations — the odd-prime multiplier calculus (M_x , primitive root g) has no qubit counterpart.

Figure 8 of Selinger, *Generators and relations for n -qubit Clifford operators* →

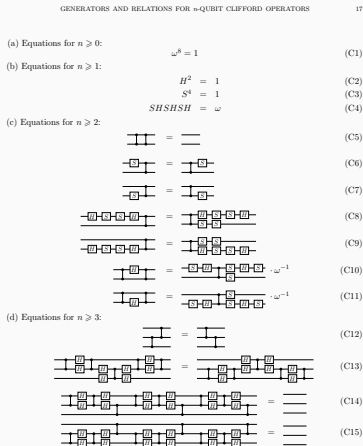


Figure 8: A presentation of the (strict spatial monoidal) Clifford groupoid by generators and relations

Composition

Composing: the Pauli factor

The projective Pauli group $(\mathbb{Z}_p \times \mathbb{Z}_p)^n$ has an easy presentation over generators X, Z on each wire:

```
data _PRel, _===_ : (n : ℕ) → WRel (Gen n) where
```

```
  order-X  : (1+ n) PRel, X ^ p === ε
```

```
  order-Z  : (1+ n) PRel, Z ^ p === ε
```

```
  comm-Z-X : (1+ n) PRel, Z • X === X • Z
```

```
Theorem-Pauli : ∀ {n} → (XZ._QRel, _===_ n) IsPresentationOf (+_p-group n)
```

```
Theorem-Pauli = XZ.presentation
```



Composing: the semidirect product

Semidirect product: relations of both factors, plus twisted commuting, $h n' = ({}^h n') h$, so $(n, h) \cdot (n', h') = (n \cdot {}^h n', h h')$:

```
conj : Sym.Gen n → XZ.Gen n → Word (XZ.Gen n)

conj Sym.H-gen XZ.X-gen      = XZ.Z
conj Sym.H-gen XZ.Z-gen      = XZ.X ^ p-1
conj Sym.S-gen XZ.X-gen      = XZ.X • XZ.Z
conj Sym.S-gen XZ.Z-gen      = XZ.Z
conj Sym.CZ-gen XZ.X-gen     = XZ.X • XZ.Z XZ.†
conj Sym.CZ-gen (XZ.X-gen XZ.†) = XZ.X XZ.† • XZ.Z
```

```
Pauli⊗Symplectic : (n : ℕ) → WRel (XZ.Gen n ⊔ Sym.Gen n)
```

```
Pauli⊗Symplectic n = XZ._QRel,===_ n ♦ Sim._QRel,===_ n ♦ ConjRelw conj
```

The projection to the symplectic factor is a homomorphism, so symplectic circuits and their relations **lift** unchanged

Plumbing: presentation simplification

Two presentations are **equivalent** when the presented groups are isomorphic over the same interpretation — each side's axioms derived from the other's. All Agda-checked, e.g. `Paper-V0.Iso, Simplified.Iso`.

	SemiDirect	Paper-V0
generators	X, Z, H, S, CZ per wire	H, S, CZ per wire
relations	Pauli + symplectic + conjugation	16
<i>generator reduction</i> : X, Z become the derived words from the introduction		
<i>relation reduction</i> : conjugation rules become derivable, e.g. $CZ \cdot X\uparrow \approx X\uparrow \cdot Z\downarrow \cdot CZ$		

Getting the scalars back

The Clifford group itself is a central extension by $\langle \omega \rangle$. On presentations: add a generator ω with $\omega^p = \varepsilon$, make it commute with every gate, and **correct** each mod-scalar relation by the scalar it actually picks up — only **order-SH** needs one:

$$\text{---} \boxed{S} \text{---} \boxed{H} \text{---} \boxed{S} \text{---} \boxed{H} \text{---} \boxed{S} \text{---} \boxed{H} \text{---} \equiv \boxed{\omega^{(p^2-1)/8} I}$$

Theorem-Clifford : $\forall (n : \mathbb{N}) \rightarrow$

$(\text{QS_Exact, __==_ } n) \text{ IsPresentationOf}$

$(\text{PE.Presented-group } n \text{ (CQP.exact-generator-data } n))$

Theorem-Clifford = **CQP.presentation-exact**

(δ_p : the exponent $(p^2 - 1)/8$ is exact division — p odd — and the correction is checked, not assumed.)

Conclusion

Conclusion

- **Main result 1:** presentations by generators and relations of the qudit Clifford operators (16 relations, ≤ 3 qudits) and of the symplectic operators, for all odd primes at once.
- **Main result 2:** a unique normal form for qudit Clifford and symplectic operators.
- Both are **checked in Agda** — except for the two classical glue facts: that projective Clifford is $\mathbf{Pauli} \rtimes \mathbf{Sp}$, and that Clifford is a central extension of projective Clifford by the scalars.

Thank you!